Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Parallel rays from a distant object fall on a solid transparent sphere of radius R and refractive index μ. The distance of the image from the sphere is

(A) $\frac{R(2-\mu)}{2(\mu-1)}$

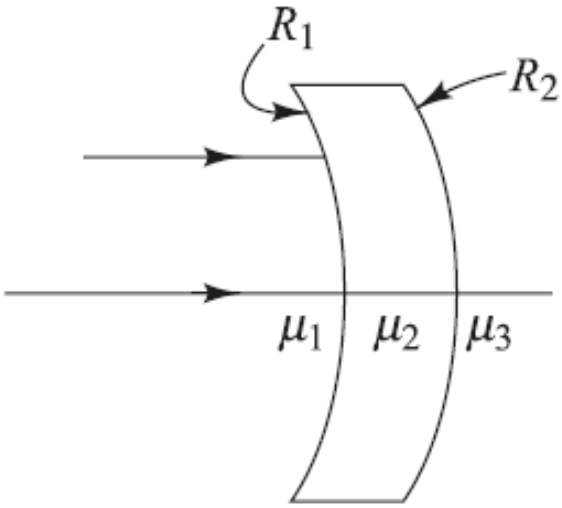
(B) $\frac{R\mu}{(\mu-1)}$

(C) $\frac{R}{(\mu-1)}$

(D) $R(\mu-1)$

Q.2

Figure shows a lens of negligible thickness having radii of curvature R_1 and R_2 and $\mu_1 < \mu_2 < \mu_3$. If $R_1 = R_2 = R$, the focal length of the lens is



(A) $f = \frac{\mu_2 R}{(\mu_1-\mu_3)}$

(B) $f = \frac{\mu_3 R}{(\mu_1-\mu_2)}$

(C) $f = \frac{\mu_1 R}{(\mu_2-\mu_3)}$

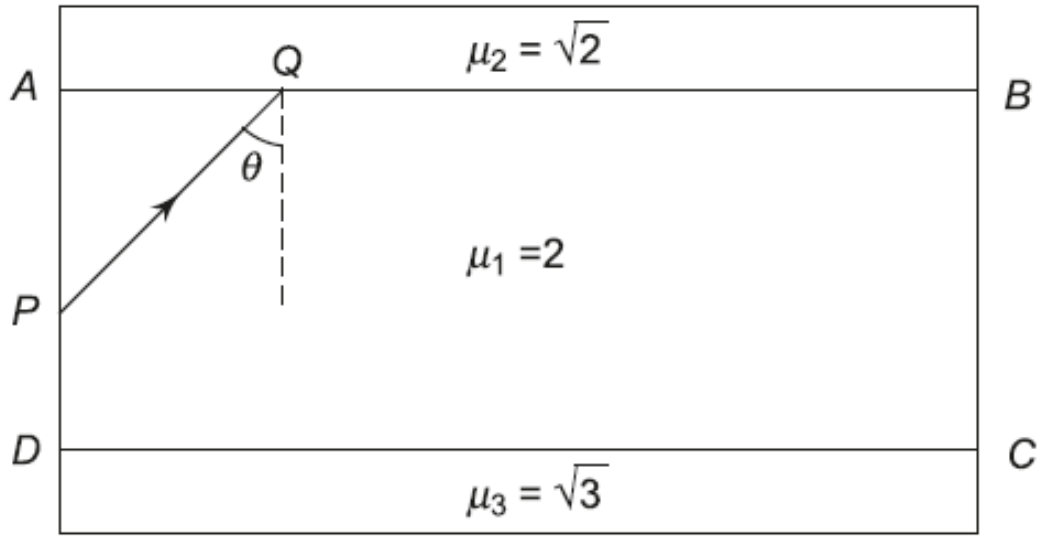
(D) $f = \frac{\mu_3 R}{(\mu_1-\mu_3)}$

Q.3

A parallel sides slab $ABCD$ of refractive index 2 is sandwiched between two slabs of refractive indices $\sqrt{2}$ and $\sqrt{3}$ as shown in the Fig. 16.50. The minimum value of angle θ such that the ray PQ suffers

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total internal reflection at both the surfaces AB and CD is



(A) 30°

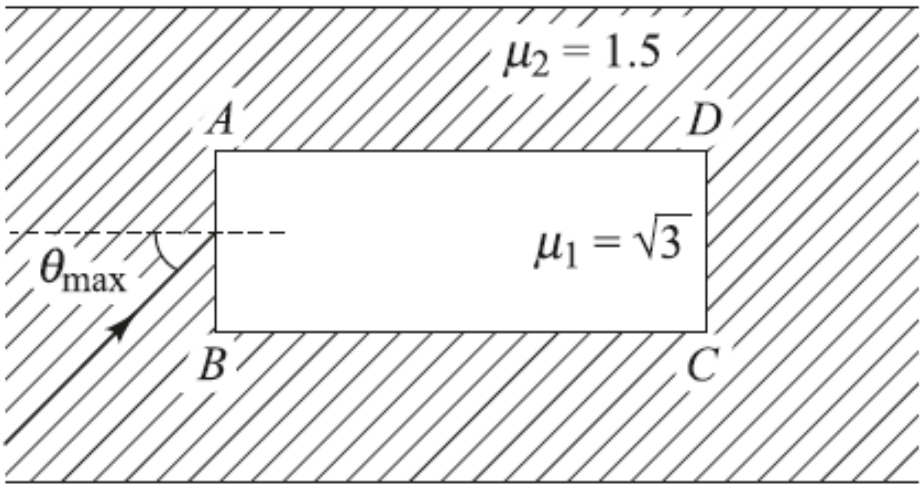
(B) 45°

(C) 60°

(D) 75°

Q.4

A rectangular glass slab $ABCD$ of refractive index $\mu_1 = \sqrt{3}$ is embedded in a transparent medium of refractive index $\mu_2 = 1.5$. A ray of light is incident on face AB of the slab as shown in Fig. 16.54. The maximum value of the angle of incidence $\theta_{\max}$ such that the ray comes out only from the other face CD is given by



(A) $\sin^{-1}\left(\frac{\sqrt{3}}{4}\right)$

(B) $\sin^{-1}\left(\frac{2}{3}\right)$

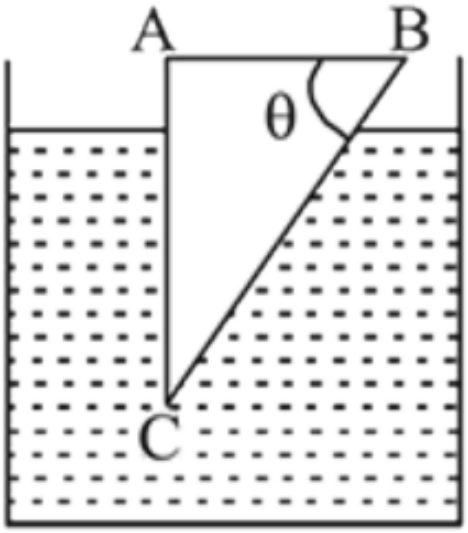
(C) $\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

(D) 60°

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Q.5

A glass prism of refractive index 1.5 is immersed in water (refractive index $4/3$). A light beam incident normally on the face AB is totally reflected to reach on the face BC if



(A) $\sin\theta \geq \frac{8}{9}$

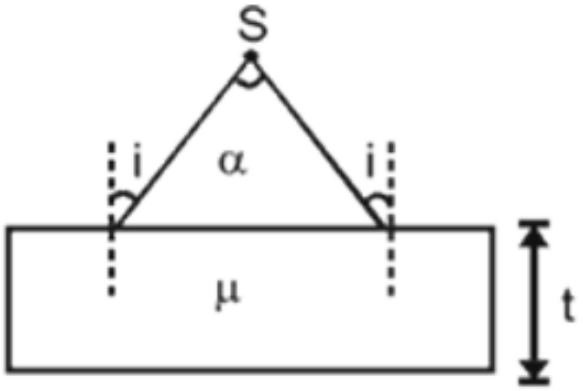
(B) $\frac{2}{3} < \sin\theta < \frac{8}{9}$

(C) $\sin\theta \leq \frac{2}{3}$

(D) $\frac{1}{2} < \sin\theta < 1$

Q.6

A diverging beam of light from a point source S having divergence angle α, falls symmetrically on a glass slab as shown. The angles of incidence of the two extreme rays are equal. If the thickness of the glass slab is t and the refractive index n, then the divergence angle of the emergent beam is



(A) zero

(B) α

(C) $\sin^{-1}\left(\frac{1}{n}\right)$

(D) $2\sin^{-1}\left(\frac{1}{n}\right)$

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Q.7

Light is incident from a medium into air at two possible angles of incidence (A) 20° and (B) 40°. In the medium light travels $3.0\ \mathrm{cm}$ in $0.2\ \mathrm{ns}$. The ray will :

(A) suffer total internal reflection in both cases (A) and (B)

(B) suffer total internal reflection in case (B) only

(C) have partial reflection and partial transmission in case (B)

(D) have 100% transmission in case (A)

Q.8

A point source of light is placed at a depth of h below the surface of water of refractive index μ. A floating opaque disc is placed on the surface of water so that light from the sources is not visible from the surface. The minimum diameter of the disc is

(A) $2h/\left(\mu^2-1\right)^{1/2}$

(B) $2h\left(\mu^2-1\right)^{1/2}$

(C) $h/\left[2\left(\mu^2-1\right)^{1/2}\right]$

(D) $h\left(\mu^2-1\right)^{1/2}$

Q.9

A prism placed in air made up of flint glass is such that any incident ray on one surface does not emerge from the second surface. Critical angle for flint glass is 36° in air. Then, refracting angle A may be

(A) 37°

(B) 54°

(C) 71°

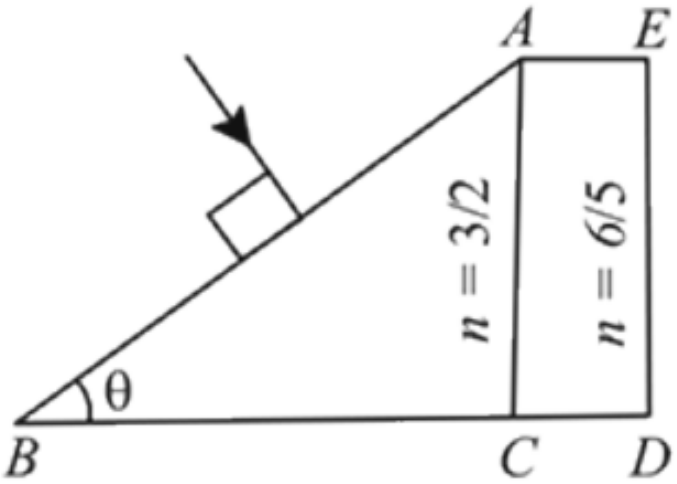
(D) 73°

Q.10

In Fig. ABC is the cross section of a right - angled prism and ACDE is the cross section of a glass slab. The value of θ so that incident normally on the face AB does not cross the face AC is (given

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$\sin^{-1}(3/5) = 37^\circ$)



(A) $\theta \leq 37^\circ$

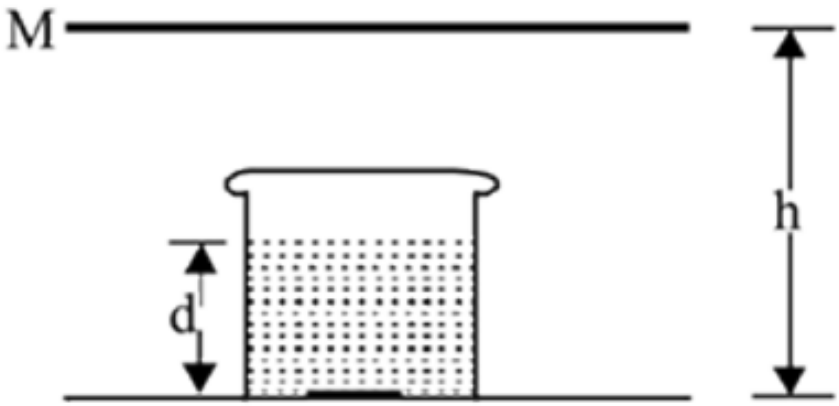
(B) $\theta < 37^\circ$

(C) $\theta \leq 53^\circ$

(D) $\theta < 53^\circ$

Q.11

A plane mirror is held at a height h above the bottom of an empty beaker. The beaker is now filled with water up to depth d. The general expression for the distance from a scratch at the bottom of the beaker to its image in terms of h and the depth d of water in the beaker is



(A) $2h - d\left(\frac{\mu}{\mu-1}\right)$

(B) $2h - \frac{d}{2}\left(\frac{\mu-1}{\mu}\right)$

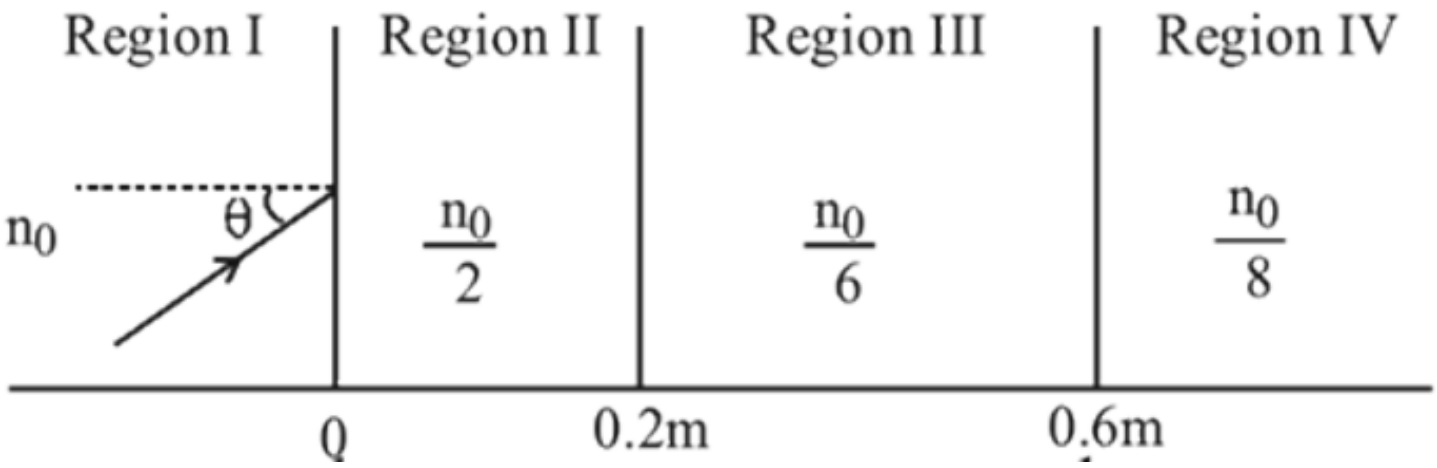
(C) $2h - d\left(\frac{\mu-1}{\mu}\right)$

(D) $2h - d\left(\frac{2\mu-1}{\mu}\right)$

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Q.12

A light beam is travelling from Region 1 to IV (figure). The refractive index in regionals I, II, III and IV are n_0, $\frac{n_0}{2}$, $\frac{n_0}{6}$ and $\frac{n_0}{8}$ respectively. The angle of incidence θ for which the beam just misses entering region IV is -



(A) $\sin^{-1}(3/4)$

(B) $\sin^{-1}(1/8)$

(C) $\sin^{-1}(1/4)$

(D) $\sin^{-1}(1/3)$

Q.13

A mango tree is at the bank of river and one of the branch of tree extends over the river. A tortoise lives in river. A mango falls just above the tortoise. The acceleration of the mango falling from tree appearing to the tortoise is (Refractive index of water is $4/3$ and the tortoise is stationary)

(A) g

(B) $3\,g/4$

(C) $4\,g/3$

(D) None of these

Q.14

A mango tree is at the bank of river and one of the branch of tree extends over the river. A tortoise lives in river. A mango falls just above the tortoise. The acceleration of the mango falling from tree appearing to the tortoise is (Refractive index of water is $4/3$ and the tortoise is stationary)

(A) g

(B) $3\,g/4$

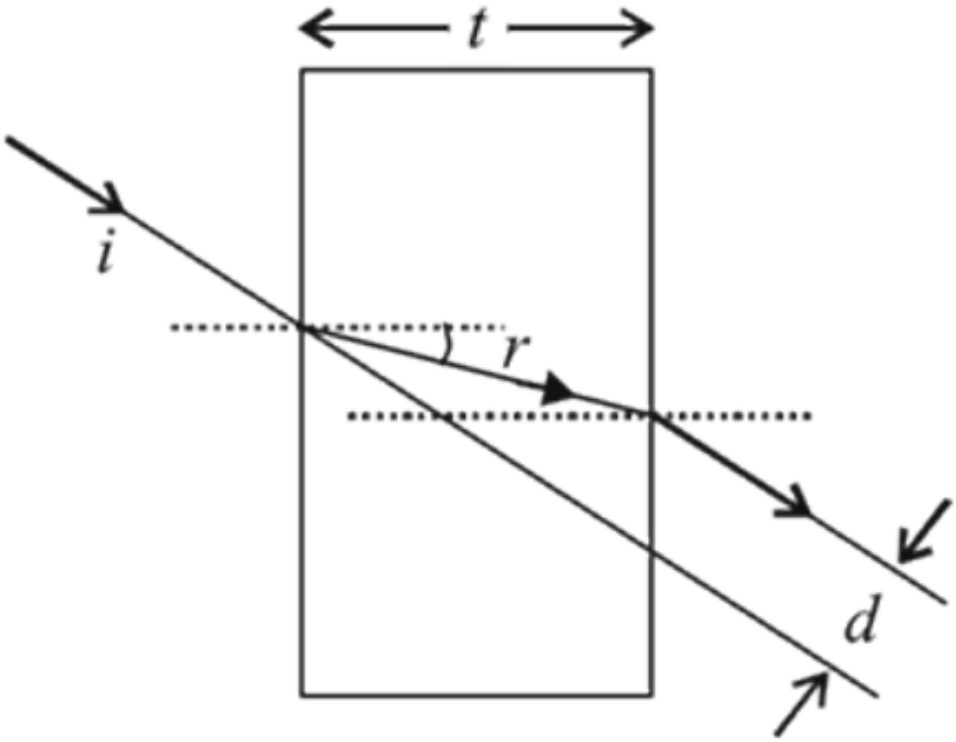
(C) $4\,g/3$

(D) None of these

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Q.15

A ray of light is incident on a thick slab of glass (thickness t) as shown below. The emergent ray is parallel to the incident ray but displaced sideways by a distance d. If the angles are small then d is:



(A) $t(1-i/r)$

(B) $rt(1-i/r)$

(C) $it\left(1-\frac{r}{i}\right)$

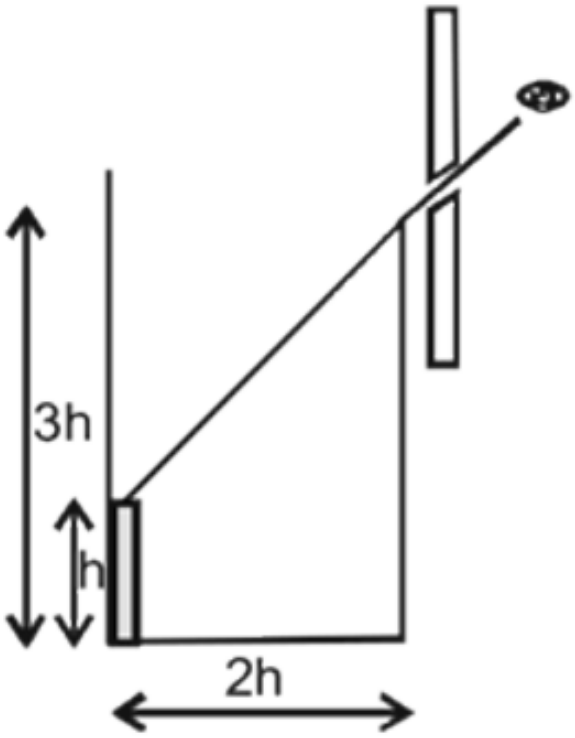
(D) $t\left(1-\frac{r}{i}\right)$

Q.16

An observer can see through pin-hole the top end of a thin rod of height h, placed as shown in the figure. The beaker height is $3\ \mathrm{h}$ and its radius h. When the beaker is filled with a liquid up to height $2\ \mathrm{h}$, he can

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see the lower end of the rod. Then the refractive index of the liquid is



(A) $\frac{5}{2}$

(B) $\sqrt{\frac{5}{2}}$

(C) $\sqrt{\frac{3}{2}}$

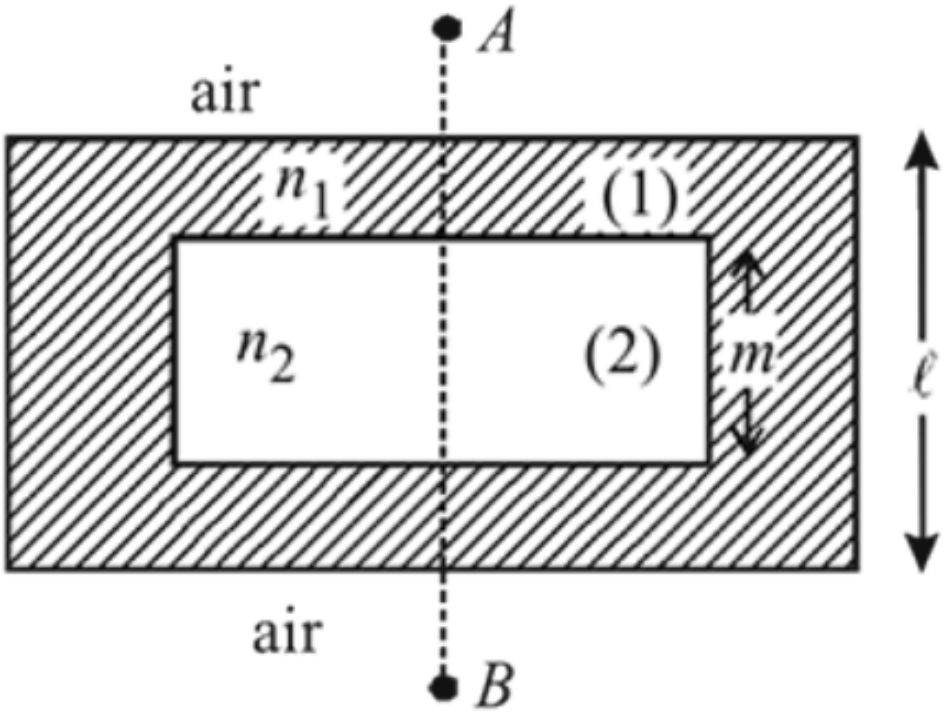
(D) $\frac{3}{2}$

Q.17

In a thick glass slab of thickness ℓ and refractive index n_1 a cuboidal cavity of thickness m is carved as shown in the figure and is filled with liquid of R.I. n_2 $(n_1 > n_2)$. The ratio of ℓ/m, so that shift

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produced by this slab is zero when an observer A observes an object B with paraxial rays is



(A) $\frac{n_1-n_2}{n_2-1}$

(B) $\frac{n_1-n_2}{n_2(n_1-1)}$

(C) $\frac{n_1-n_2}{n_1-1}$

(D) $\frac{n_1-n_2}{n_1(n_2-1)}$

Q.18

A printed page is pressed by a glass of water. The refractive index of the glass and water is 1.5 and 1.33, respectively. If the thickness of the bottom of glass is 1 cm and depth of water is 5 cm, how much the page will appear to be shifted if viewed from the top?

(A) 1.033 cm

(B) 3.581 cm

(C) 1.533 cm

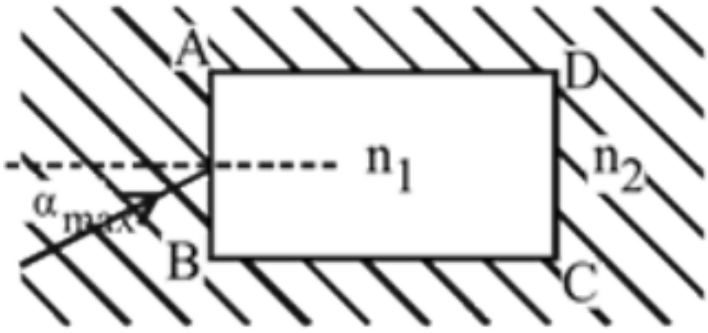
(D) 1.90 cm

Q.19

A rectangular glass slab $ABCD$ of refractive index n_1 is immersed in water of $=$
n_2). A ray of light is incident at the surface AB of the slab as shown. The maximum value of the angle

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of incidence α_{max} such that the ray comes out only from the other surface CD is given by



(A) $\sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\left(\frac{n_2}{n_1}\right)\right)\right]$

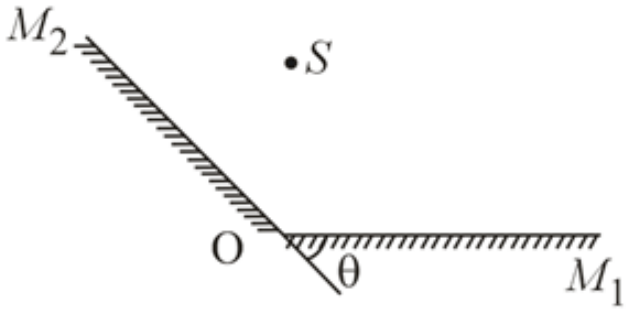
(B) $\sin^{-1}\left[n_1\cos\left(\sin^{-1}\left(\frac{1}{n_2}\right)\right)\right]$

(C) $\sin^{-1}\left(\frac{n_1}{n_2}\right)$

(D) $\sin^{-1}\left(\frac{n_2}{n_1}\right)$

Q.20

In the arrangement shown in figure, S is a point source, M_1 and M_2 are two plane fixed mirrors inclined at an angle θ with each other. If $OS = b$, then the value of distance between two images formed.



(A) $b\ \sin\ 2\theta$

(B) $2b\ \sin\ \theta$

(C) $2b\ \sin\ 2\theta$

(D) $b\ \sin\ \theta$

Answers & Solutions

Q.1 Answer:

$\frac{R(2-\mu)}{2(\mu-1)}$

Solution:

Refer to Fig. For refraction at face I.

$\frac{1}{u_1}+\frac{\mu}{v_1}=\frac{\mu-1}{R}$

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Since $u_1 = \infty$, we have

$\frac{\mu}{v_1} = \frac{\mu-1}{R}$

or $v_1 = \frac{\mu R}{(\mu-1)}$ **(1)**

For refraction at face II, $u_2 = -(v_1 - 2R) = 2R - v_1$**. Using Eq. (1), we get**

$u_2 = 2R - \frac{\mu R}{(\mu-1)} = \frac{R(\mu-2)}{(\mu-1)}$ **(2)**

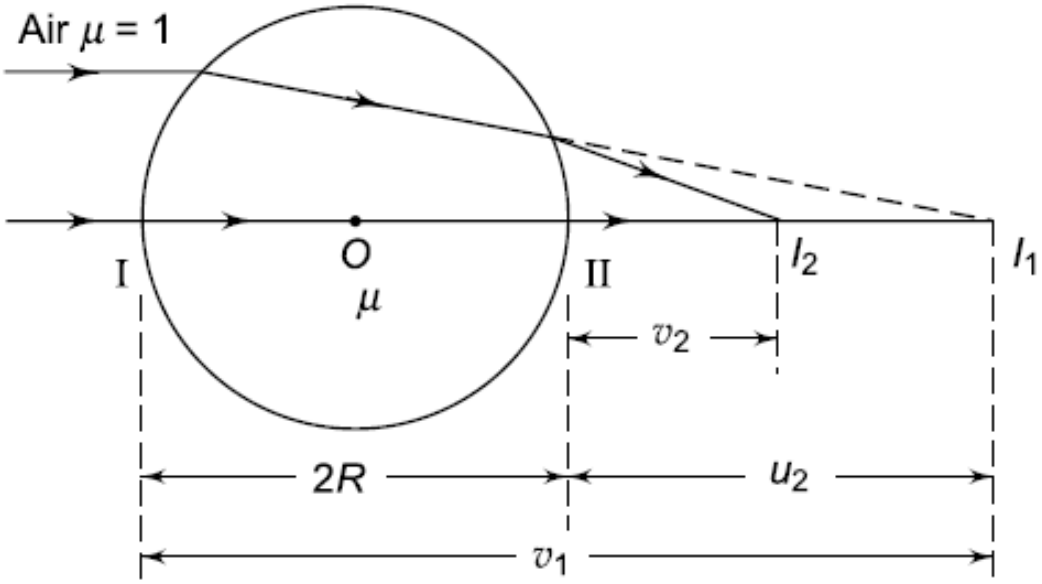
The image distance v_2 **is given by**

$\frac{\mu}{u_2} + \frac{1}{v_2} = \frac{\mu-1}{R}$ **(3)**

Using Eq. (2) in Eq. (3) and simplifying, we get

$v_2 = \frac{R(2-\mu)}{2(\mu-1)}$.

The correct choice is (a).



Q.2 Answer:

$f = \frac{\mu_3 R}{(\mu_1 - \mu_3)}$

Solution:

Refer to Fig.

I' **is the image due to refraction at the first face. For refraction at the first face, we have**

$$\frac{\mu_2}{v'} - \frac{\mu_1}{-\infty} = \frac{\mu_2 - \mu_1}{R_1}$$

$$\Rightarrow \quad \frac{\mu_2}{v'} = \frac{\mu_2 - \mu_1}{R_1} \quad (1)$$

Image I' **serves as the object for refraction at the second face and** I **is the final image. For this refraction,** $u = v' - t = v'(\because t$ **is negligible)**

$\frac{\mu_3}{v} - \frac{\mu_2}{v'} = \frac{\mu_3 - \mu_2}{R_2}$

Since the incident rays are parallel, $v = f$.

$\because \frac{\mu_3}{f} - \frac{\mu_2}{v'} = \frac{\mu_3 - \mu_2}{R_2}$ **(2)**

Adding (1) and (2), we get

$\frac{1}{f} = \left(\frac{\mu_2 - \mu_1}{\mu_3}\right)\frac{1}{R_1} + \left(\frac{\mu_3 - \mu_2}{\mu_3}\right)\frac{1}{R_2}$ **(3)**

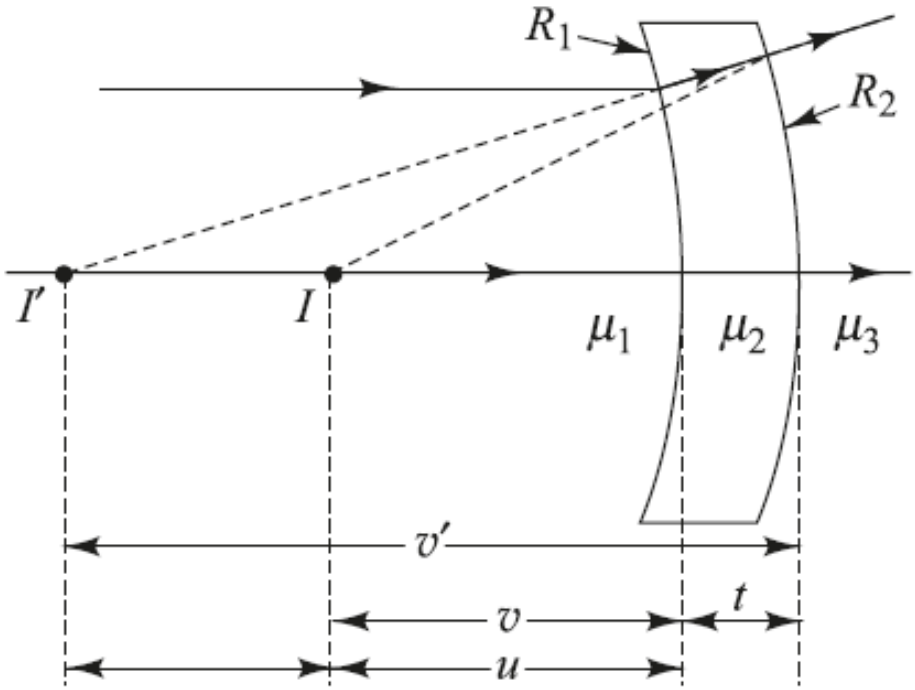
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Equation (3) is the general expression for the focal length of the given lens. It holds for both convex and concave lenses and has sign conventions. It is given that $R_1 = R_2 = -R$. Putting this in Eq. (3), we get

$$\frac{1}{f} = -\frac{(\mu_3 - \mu_1)}{\mu_3 R}$$

$$\Rightarrow \quad f = -\frac{\mu_3 R}{(\mu_3 - \mu_1)}$$

The negative sign shows that the given lens is a concave (diverging) lens. So the correct choice is (d).



Q.3 Answer:

60°

Solution:

Refer to Fig.

For total internal reflection at surface AB, angle θ must be greater than or equal to the critical angle i_1 given by

$\sin i_1 = \frac{\mu_2}{\mu_1} = \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}}$

which gives $i_1 = 45^\circ$

For total internal reflection at surface CD, angle θ must be greater than or equal to the critical angle i_2 given by

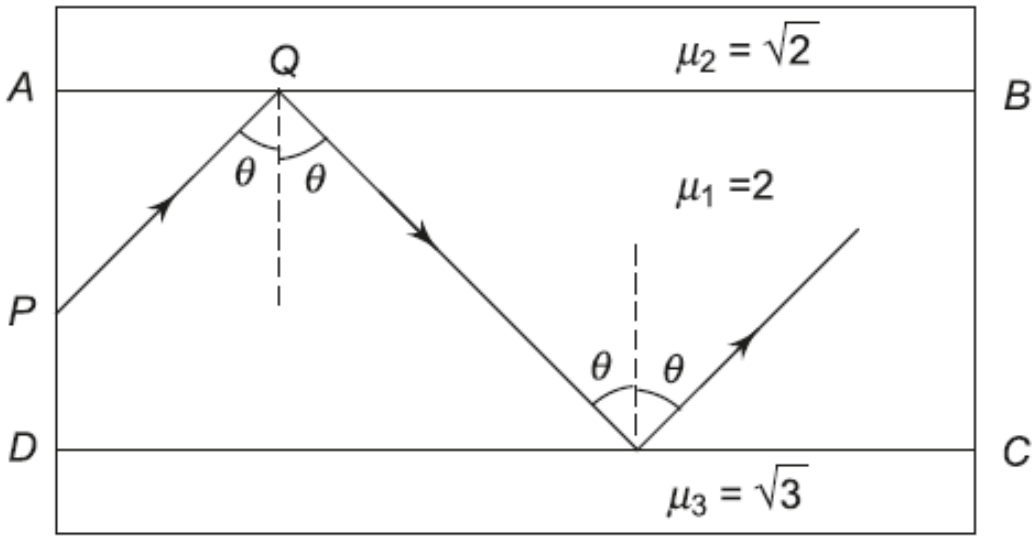
$\sin i_2 = \frac{\mu_3}{\mu_1} = \frac{\sqrt{3}}{2}$

which gives $i_2 = 60^\circ$.

Hence, for total internal reflection at both the surfaces AB and CD, the minimum value of

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$\theta = 60^\circ$. **Thus the correct choice is** (c)



Q.4 Answer:

$\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$

Solution:

The ray will emerge from side CD **if the ray incident on face** AB **after entering the slab suffers repeated total internal reflections at faces** AD **and** BC **of the slab as shown in Fig.**

From Snell's law,

$$\mu_2 \sin\theta_{\max} = \mu_1 \sin r$$

$$\Rightarrow \quad \sin\theta_{\max} = \frac{\mu_1}{\mu_2}\sin r \quad (1)$$

The critical angle is given by

$\sin i_c = \frac{\mu_2}{\mu_1}$

For total internal reflection, angle i **must at least be equal** i_c**. Therefore,** $r = 90^\circ - i_c$**. Using this (1) we get**

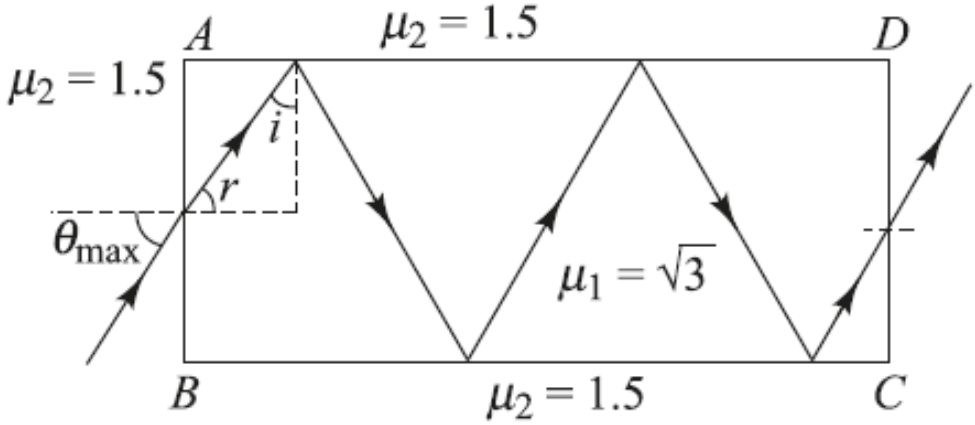
$\sin\theta_{\max} = \frac{\mu_1}{\mu_2}\sin(90^\circ - i_c) = \frac{\mu_1}{\mu_2}\cos i_c$ **(2)**

Now $\sin i_c = \frac{\mu_2}{\mu_1} = \frac{1.5}{\sqrt{3}} = \frac{\sqrt{3}}{2} \Rightarrow i_c = 60^\circ$**. Therefore,**

$\cos i_c = \cos 60^\circ = \frac{1}{2}$**. Using this in (2) we get**

$\sin\theta_{\max} = \frac{\sqrt{3}}{1.5} \times \frac{1}{2} = \frac{1}{\sqrt{3}}$

So the correct choice is (c)**.**



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Q.5 Answer:

$\sin\theta \geq \frac{8}{9}$

Solution:

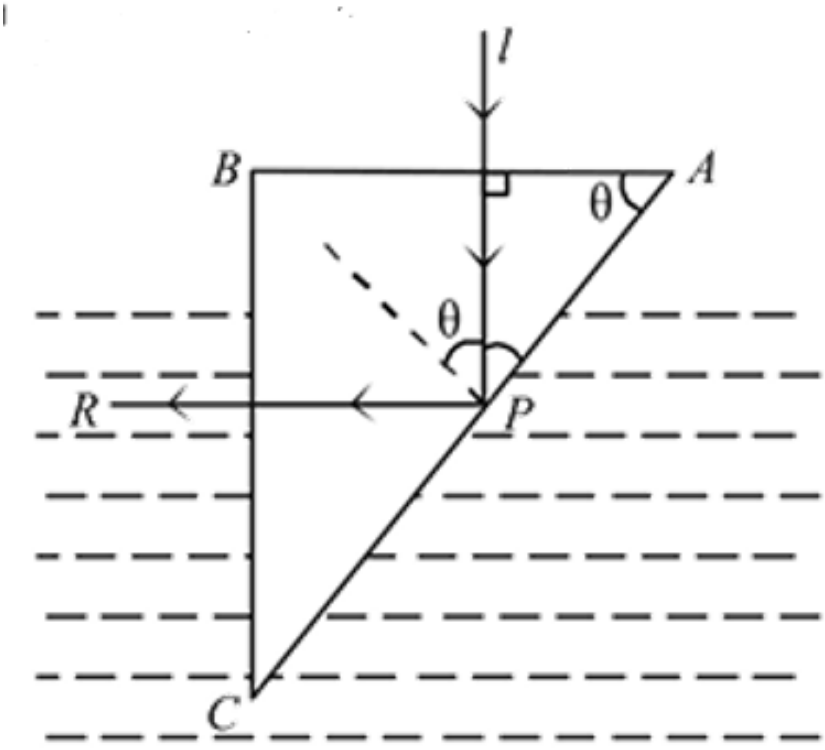
(a) The phenomenon of total internal reflection takes place during reflection at P.

$\sin\theta = \frac{1}{{}_{g}\mu} \quad \ldots(\mathrm{i})$

Now, ${}^{w}_{g}\mu = \frac{{}^{a}_{}\mu}{{}^{a}_{w}\mu} = \frac{1.5}{4/3} = 1.125$

$\therefore \sin\theta = \frac{1}{1.125} = \frac{8}{9}$

$\therefore$ **$\sin\theta$ should be greater than $\frac{8}{9}$**



Q.6 Answer:

α

Solution:

No solution available

Q.7 Answer:

suffer total internal reflection in case (B) only

Solution:

No solution available

Q.8 Answer:

$2h/\left(\mu^2-1\right)^{1/2}$

Solution:

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The figure shows incidence from water at critical angle θ_c for the limiting case.

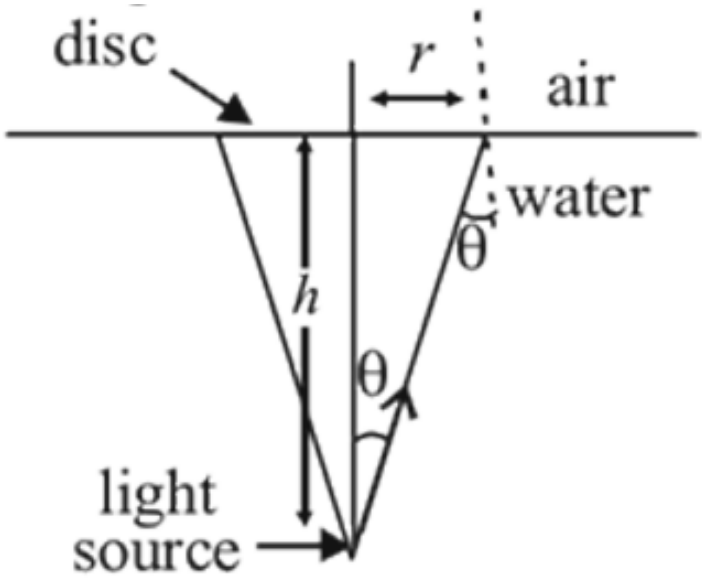
Now, $\sin\theta_c = 1/\mu$ so that

$\tan\theta_c = 1/\left(\mu^2 - 1\right)^{1/2}$

which is also equal to r/h where r is the radius

of the disc.

Therefore, diameter of the disc is $2r = 2h\tan\theta_c$.



Q.9 Answer:

73°

Solution:

In the prism the to occur total internal reflection the refracting angle a must be more than twice the critical angle of the prism material. In this case, critical angle for flint glass is 36 so the refracting angle. It must be greater than twice of the critical angle that is it must be greater than 72.

Q.10 Answer:

$\theta < 37^\circ$

Solution:

$\mathrm{A} = 90^\circ - \theta$

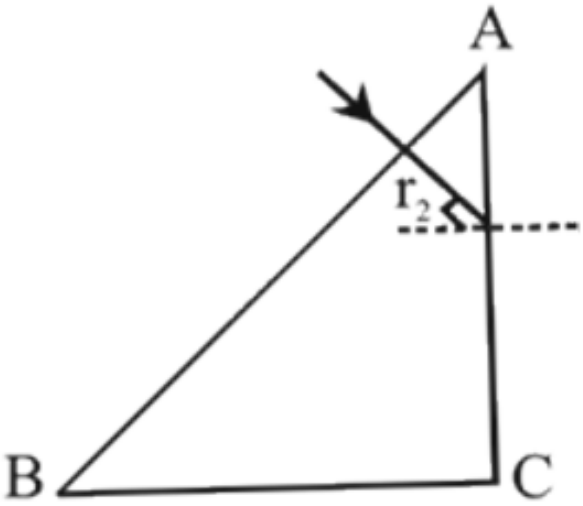
$\Rightarrow \quad r_2 = A = 90^\circ - \theta > \theta_c$

$\cos\theta > \sin\theta_c = \frac{6/5}{2/3} = \frac{4}{5}$

(θ_c **is critical angle)**

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$\theta < \cos^{-1}\frac{4}{5} = 37^\circ$



Q.11 Answer:

$2h - d\left(\frac{\mu-1}{\mu}\right)$

Solution:

shift observed is $\mathrm{d}\left(1-\frac{1}{\mu}\right)$

object distance wrt mirror is $\mathrm{h}-\text{shift}=\mathrm{h}-\mathrm{d}\left(1-\frac{1}{\mu}\right)$

image distance from mirror is $\mathrm{h}-\mathrm{d}\left(1-\frac{1}{\mu}\right)$

distance of image from the scratch at the bottom of beaker is $\mathrm{h}-\mathrm{d}\left(1-\frac{1}{\mu}\right)+\mathrm{h}=2\,\mathrm{h}-\mathrm{d}\left(\frac{\mu-1}{\mu}\right)$

option C **is correct**

Q.12 Answer:

$\sin^{-1}(1/8)$

Solution:

No solution available

Q.13 Answer:

$4\ \mathrm{g}/3$

Solution:

$\frac{x}{1}=\frac{x_{rel}}{\mu} \Rightarrow x_{rel}=\mu x$

$\Rightarrow \frac{d^2x_{rel}}{dt^2}=\mu^2\frac{d^2x}{dt^2} \Rightarrow a_{rel}=\mu \mathrm{g}$

Q.14 Answer:

$4\ \mathrm{g}/3$

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Solution:

$\frac{x}{1} = \frac{x_{rel}}{\mu} \Rightarrow x_{rel} = \mu x$
$\Rightarrow \frac{d^2 x_{rel}}{dt^2} = \mu^2 \frac{d^2 x}{dt^2} \Rightarrow a_{rel} = \mu g$

Q.15 Answer:

$it\left(1 - \frac{r}{i}\right)$

Solution:

The lateral displacement is given by $d = \frac{t}{\cos r} \sin(i - r)$
For small angle i**, angle** r **is also small, and so** $d = t(i - r) = ti(1 - r/i)$

Q.16 Answer:

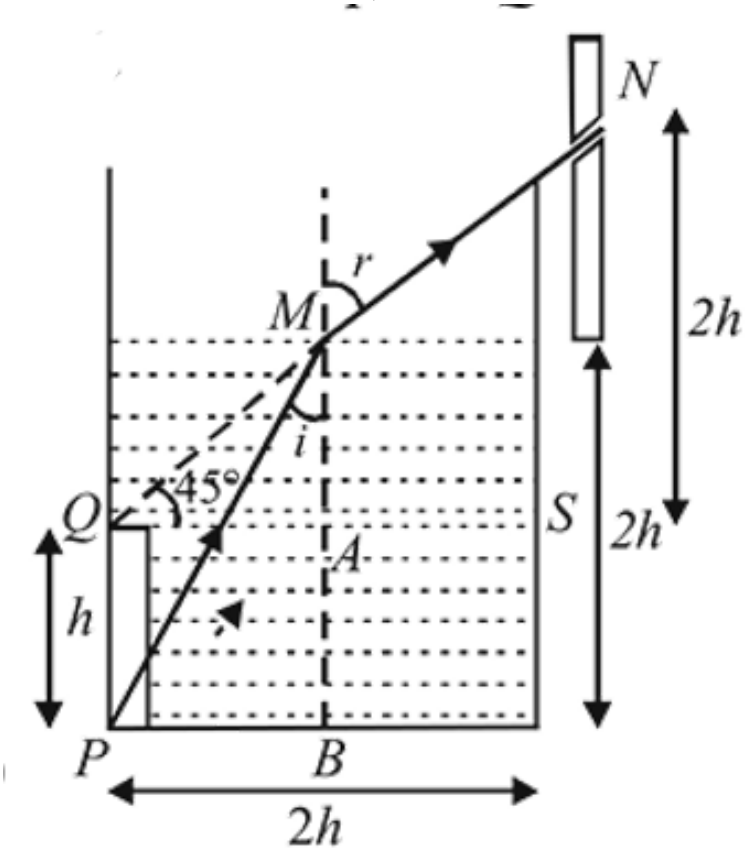
$\sqrt{\frac{5}{2}}$

Solution:

For the image of point P **to be seen by the observer, it should be formed at point** Q**.**
In $\Delta QNS, NS = QS = 2h$
$\therefore \quad \angle NQS = 45^\circ$
$\therefore \quad r = 45^\circ$
Now in ΔQMA**,**
$\angle MQA = 45^\circ$
$\therefore \quad MA = QA = h$
$\mu = \frac{\sin r}{\sin i}$
$^2 = \frac{\sin 45^\circ}{\sin \mathrm{i}} \ldots (i)$
In $\triangle PMB$
$PM^2 = 4h^2 + h^2 = 5h^2$
$\sin i = \frac{h}{\sqrt{5}h} = \frac{1}{\sqrt{5}}$**...(ii)**

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From (i) and (ii) $_1^2\mu = \sqrt{\frac{5}{2}}$



Q.17 Answer:

$\frac{n_1 - n_2}{n_2(n_1 - 1)}$

Solution:

$$\textbf{Shift} = (\ell - m)\left(1 - \frac{1}{n_1}\right) + m\left(1 - \frac{1}{n_2}\right) = 0$$

Q.18 Answer:

1.533 cm

Solution:

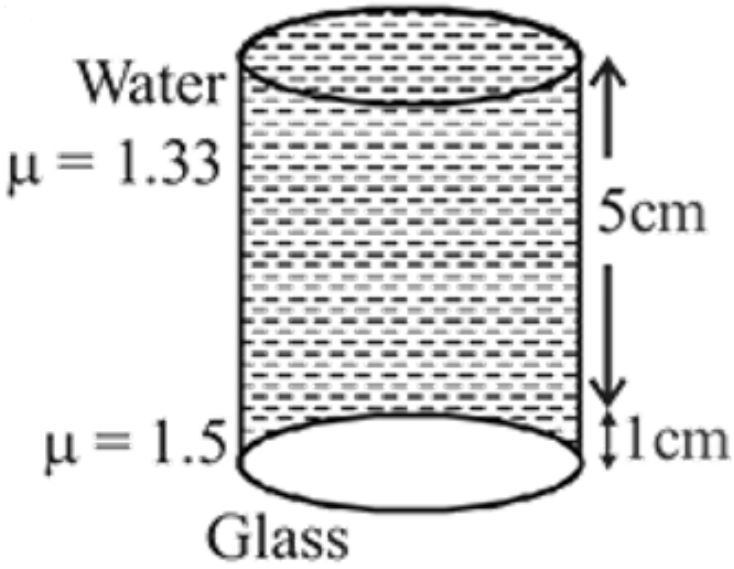
$$\text{(c) Real depth } = 5 \text{ cm} + 1 \text{ cm} = 6 \text{ cm}$$
$$\text{Apparent depth } = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \ldots$$

$= \frac{5}{1.33} + \frac{1}{1.5} \simeq 3.8 + 0.7$
$= 4.5$ cm

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$\therefore$ Shift $= 6\text{ cm} - 4.5\text{ cm}$

$\therefore$ $\cong 1.5\text{ cm}$



Q.19 Answer:

$\sin^{-1}\left[\frac{n_1}{n_2}\cos\left(\sin^{-1}\left(\frac{n_2}{n_1}\right)\right)\right]$

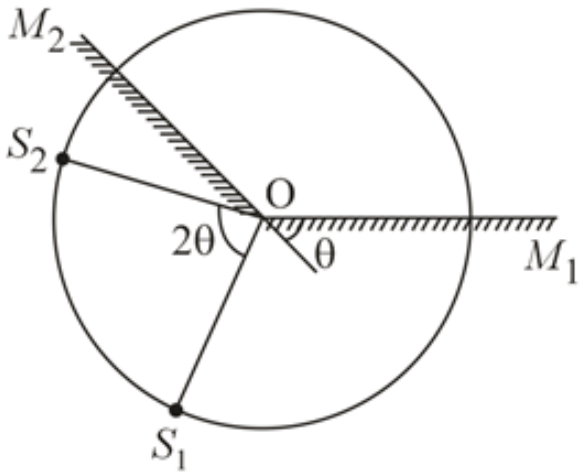
Solution:

No solution available

Q.20 Answer:

$2b \quad \sin \quad \theta$

Solution:



If mirror rotates by angle θ image of object rotates by angle 2θ.

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